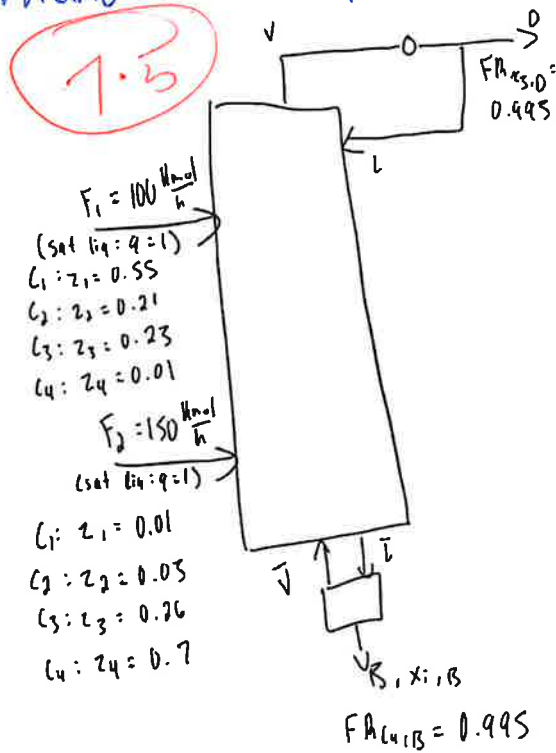


D5. A distillation column with a total condenser and a partial reboiler is separating two feeds. Feed 1 is a saturated liquid, and its rate is 100.0 kmol/h. Feed 1 is 55.0 mol% methanol, 21.0 mol% ethanol, 23.0 mol% propanol, and 1.0 mol% butanol. Feed 2 is a saturated liquid with a flow rate of 150.0 kmol/h. Feed 2 is 1.0 mol% methanol, 3.0 mol% ethanol, 26.0 mol% propanol, and 70.0 mol% butanol. We want to recover 99.3% of the propanol in distillate and 99.5% of the butanol in bottoms. Make appropriate assumptions, and find distillate and bottoms flow rates and mole fractions of distillate and bottoms.



- ① C_1 - LNK (volatility: $C_1 > C_2 > C_3 > C_4$)
 C_2 - LNK (bp: $C_1 < C_2 < C_3 < C_4$)
 C_3 - LH } targets
 C_4 - HH

- ② assume: - all LNK collects in distillate
 $FR_{C_1,D} = 1 \rightarrow x_{C_1,D} \cdot D = F_1 \cdot z_{C_1,1} \rightarrow x_{C_1,D} \cdot D = 56.5$
 $FR_{C_2,D} = 1 \rightarrow x_{C_2,D} \cdot D = F_1 \cdot z_{C_2,1} + F_2 \cdot z_{C_2,2} \rightarrow x_{C_2,D} \cdot D = 25.5$

③ Distillate

$$C_1: D \cdot x_{C_1,D} = F_1 \cdot z_{C_1,1} + F_2 \cdot z_{C_1,2} = (100 \frac{\text{kmol}}{\text{h}})(0.55) + (150 \frac{\text{kmol}}{\text{h}})(0.01) = 56.5 \frac{\text{kmol}}{\text{h}}$$

$$C_2: D \cdot x_{C_2,D} = F_1 \cdot z_{C_2,1} + F_2 \cdot z_{C_2,2} = (100)(0.21) + (150)(0.03) = 25.5 \frac{\text{kmol}}{\text{h}}$$

$$C_3: FR_{C_3,D} = \frac{D \cdot x_{C_3,D}}{F_1 \cdot z_{C_3,1} + F_2 \cdot z_{C_3,2}} = 0.993 \rightarrow D \cdot x_{C_3,D} = 0.993(100 \cdot 0.23 + 150 \cdot 0.26) = 61.556 \frac{\text{kmol}}{\text{h}}$$

$$C_4: FR_{C_4,D} = 1 - FR_{C_4,B} = 0.005 = \frac{D \cdot x_{C_4,D}}{F_1 \cdot z_{C_4,1} + F_2 \cdot z_{C_4,2}} \rightarrow D \cdot x_{C_4,D} = 0.005(100 \cdot 0.01 + 150 \cdot 0.7) = 0.53 \frac{\text{kmol}}{\text{h}}$$

$$D = D(x_{C_1,D} + x_{C_2,D} + x_{C_3,D} + x_{C_4,D}) \cdot x_{i,D} = \frac{D \cdot x_{i,D}}{D} \left\{ \begin{array}{l} x_{C_1,D} = \frac{D \cdot x_{C_1,D}}{D} = \frac{56.5}{144.096} = 0.39 \\ x_{C_2,D} = \frac{D \cdot x_{C_2,D}}{D} = 0.18 \\ x_{C_3,D} = \frac{D \cdot x_{C_3,D}}{D} = 0.43 \\ x_{C_4,D} = \frac{D \cdot x_{C_4,D}}{D} = 0.004 \end{array} \right.$$

$$D = 56.5 + 25.5 + 61.566 + 0.53 = 144.096 \frac{\text{kmol}}{\text{h}} \cdot x_{i,D} = 1$$

④ Bottom

$$C_1: FR_{C_1,D} = 1 \rightarrow FR_{C_1,B} = 0 \rightarrow 0 \frac{\text{kmol}}{\text{h}} = B \cdot x_{C_1,B}$$

$$C_2: FR_{C_2,D} = 1 \rightarrow FR_{C_2,B} = 0 \rightarrow 0 \frac{\text{kmol}}{\text{h}} = B \cdot x_{C_2,B}$$

$$C_3: B \cdot x_{C_3,B} = (1 - FR_{C_3,D}) (F_1 \cdot z_{C_3,1} + F_2 \cdot z_{C_3,2}) = 0.007(100 \cdot 0.23 + 150 \cdot 0.26) = 0.434 \frac{\text{kmol}}{\text{h}}$$

$$C_4: B \cdot x_{C_4,B} = FR_{C_4,B} (F_1 \cdot z_{C_4,1} + F_2 \cdot z_{C_4,2}) = 0.995(100 \cdot 0.01 + 150 \cdot 0.7) = 105.47 \frac{\text{kmol}}{\text{h}}$$

$$B = 0 + 0 + 0.434 + 105.47 = 105.904 \frac{\text{kmol}}{\text{h}} \cdot x_{i,B} = 1$$

$$x_{i,B} = \frac{B \cdot x_{i,B}}{B} \quad x_{C_1,B} = \frac{0}{B} = 0 \quad x_{C_2,B} = \frac{0}{B} = 0$$

$$x_{C_3,B} = \frac{0.434}{105.904} = 0.004 \quad x_{C_4,B} = \frac{105.47}{105.904} = 0.996$$

D2. We are separating a mixture of propylene, propane, and isobutane in a distillation column with a partial condenser and a partial reboiler at a pressure of 15.0 bar. We desire a 0.999 fractional recovery of propylene in the distillate, at least a 0.950 fractional recovery of propane in the bottoms, and at least a 0.9999 fractional recovery of isobutane in the bottoms. How many stages are required at total reflux?

Data: At 34°C, K Propylene = 1.00, K propane = 0.89, K isobutane = 0.42.

* C_3H_6 (A) * C_3H_8 (B) * C_4H_{10} (C)

At 40°C, K propylene = 1.13, K propane = 1.000, and K isobutane = 0.46.

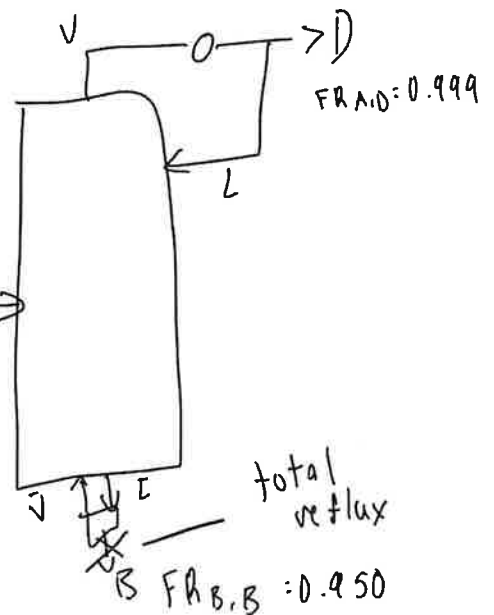
①
 C_3H_6 LH
 C_3H_8 NH
 C_4H_{10} HH

② $T_1 \propto \alpha_{AC,D} = \frac{u_{A,T_1}}{u_{B,T_1}} = \frac{1}{0.42} = 2.38$

$T_2 \propto \alpha_{AC,B} = \frac{u_{A,T_2}}{u_{B,T_2}} = \frac{1.13}{0.46} = 2.46$

$T_1 = 34^\circ C$
 $u_A = 1$
 $u_B = 0.89$
 $u_C = 0.42$

$T_2 = 40^\circ C$
 $u_A = 1.13$
 $u_B = 1.000$
 $u_C = 0.46$



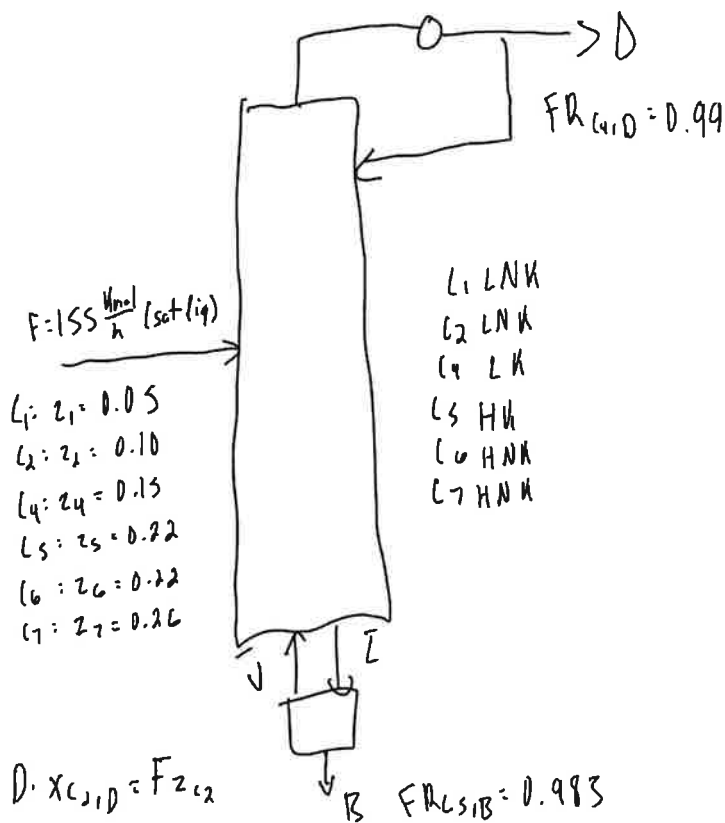
$$\alpha_{AC} = (\alpha_{AC,D} \cdot \alpha_{AC,B})^{1/2} = (2.38 + 2.46)^{1/2} = 2.42$$

$$N_{min} = \frac{\ln \left[\left(\frac{FRA_{A,D}}{1-FRA_{A,D}} \right) / \left(\frac{1-FR_{C,D}}{FR_{C,D}} \right) \right]}{\ln \alpha_{AC}} = \frac{\ln \left[\left(\frac{0.999}{1-0.999} \right) / \left(\frac{1-0.999}{0.999} \right) \right]}{\ln (2.42)}$$

$N_{min} = 15.6$

D7. Your boss wants some idea of how expensive it will be to distill 155.0 kmol/h of a saturated liquid feed that is 5.0 mol% methane, 10.0 mol% ethane, 15.0 mol% n-butane, 22.0 mol% n-pentane, 22.0 mol% n-hexane, and 26.0 mol% n-heptane. Column pressure is 700.0 kPa. The column has a partial condenser and a partial reboiler. We want to recover 99.0% of the n-butane in the distillate and 98.3% of the n-pentane in the bottoms. Do the calculations of the K values either from the DePriester chart or from Eq. (2-28).

- Assuming that NKS do not distribute, calculate the values of D and B in kmol/h and the mole fractions in distillate and bottoms.
- Do a bubble-point calculation at the feed conditions. Calculate the relative volatilities of all components with respect to the HK (n-pentane). Use these values as the average value of relative volatility for the entire column. Also determine the bubble-point temperature of the distillate to see if condensation will be expensive.
- Determine the minimum number of stages for this separation with the Fenske equation.
- Determine the minimum reflux ratio, $(L/D)_{min}$, with the Underwood method.
- Estimate the number of stages required if $L/D \approx M \times (L/D)_{min}$ with the Gilliland correlation (Davis's fit is convenient) where $M \approx 1.04, 1.10$, and 2.0 .
- Will this distillation be reasonably economical, or should an alternative be found? Briefly explain your reasoning.



a) assume: - all LNH collects in distillate

$$FR_{L1,D} = 1 \quad FR_{L2,D} = 1 \rightarrow D \cdot x_{L1,D} = F \cdot z_{L1}, \quad D \cdot x_{L2,D} = F \cdot z_{L2}$$

- all HNH collects in bottom

$$FR_{L6,B} = 1 \quad FR_{L7,B} = 1 \rightarrow B \cdot x_{L6,B} = F \cdot z_{L6}, \quad B \cdot x_{L7,B} = F \cdot z_{L7}$$

Distillate:

$$L1: D \cdot x_{L1,D} = (155)(0.05) = 7.75 \frac{\text{kmol}}{\text{h}}$$

$$L2: D \cdot x_{L2,D} = (155)(0.1) = 15.5 \frac{\text{kmol}}{\text{h}}$$

$$L4: FR_{L4,D} = 0.99 = \frac{D \cdot x_{L4,D}}{F \cdot z_{L4}} \rightarrow D \cdot x_{L4,D} = 0.99(155 \cdot 0.15) = 23.0175 \frac{\text{kmol}}{\text{h}}$$

$$L5: FR_{L5,D} = (1 - 0.983) = \frac{D \cdot x_{L5,D}}{F \cdot z_{L5}} \rightarrow D \cdot x_{L5,D} = (1 - 0.983)(155 \cdot 0.22) = 0.5797 \frac{\text{kmol}}{\text{h}}$$

$$L6: D \cdot x_{L6,D} = 0 \quad [FR_{L6,D} = 0] \quad L7: D \cdot x_{L7,D} = 0 \quad [FR_{L7,D} = 0]$$

$$D = 7.75 + 15.5 + 23.0175 + 0.5797 = 46.8472 \frac{\text{kmol}}{\text{h}}$$

$$\text{Bottoms: } L1: B \cdot x_{L1,B} = 0 \quad [FR_{L1,B} = 0] \quad L2: B \cdot x_{L2,B} = 0 \quad [FR_{L2,B} = 0]$$

$$L4: FR_{L4,B} = (1 - 0.99) = \frac{B \cdot x_{L4,B}}{F \cdot z_{L4}} \rightarrow B \cdot x_{L4,B} = (1 - 0.99)(155 \cdot 0.15) = 0.2325 \frac{\text{kmol}}{\text{h}}$$

$$L5: FR_{L5,B} = 0.983 = \frac{B \cdot x_{L5,B}}{F \cdot z_{L5}} \rightarrow B \cdot x_{L5,B} = 0.983(155 \cdot 0.22) = 33.5203 \frac{\text{kmol}}{\text{h}}$$

$$L6: B \cdot x_{L6,B} = (155 \cdot 0.22) = 34.1 \frac{\text{kmol}}{\text{h}} \quad L7: B \cdot x_{L7,B} = (155 \cdot 0.26) = 40.3 \frac{\text{kmol}}{\text{h}}$$

$$B = 0.2325 + 33.5203 + 34.1 + 40.3 = 108.1528 \frac{\text{kmol}}{\text{h}} \quad \left\{ \begin{array}{l} *x_{i,D} = \frac{D \cdot x_{i,D}}{D} \\ *x_{i,B} = \frac{B \cdot x_{i,B}}{B} \end{array} \right.$$

$$0.2325 = 1.1112 \quad x_{L4,B} = \frac{0.2325}{108.1528} = 0.00215 \quad x_{L5,B} = \frac{33.5203}{108.1528} = 0.3098 \quad x_{L6,B} = \frac{34.1}{108.1528} = 0.3153 \quad x_{L7,B} = \frac{40.3}{108.1528} = 0.3726$$

$$\begin{aligned} *x_{i,D} &= \frac{D \cdot x_{i,D}}{D} \\ x_{L1,D} &= \frac{7.75}{46.8472} = 0.165 \\ x_{L2,D} &= \frac{15.5}{46.8472} = 0.33 \\ x_{L4,D} &= \frac{23.0175}{46.8472} = 0.491 \\ x_{L5,D} &= \frac{0.5797}{46.8472} = 0.0124 \\ x_{L6,D} &= x_{L7,D} = 0 \end{aligned}$$

b) @ $T_{feed} = \sim 136^\circ C$

$$I = \sum \frac{x_{i,D}}{u_i(T)} = 26^\circ C$$

$u_{c1} = 9.4 \quad \alpha_{c1-cs} = 42.7$

$u_{c2} = 7.9 \quad \alpha_{c2-cs} = 36$

$u_{c4} = 2.3 \quad \alpha_{c4-cs} = 10.5$

$u_{cs} = 0.22$

$u_{c6} = 0.14 \quad \alpha_{c6-cs} = 0.64$

$u_{c7} = 0.09 \quad \alpha_{c7-cs} = 0.41$

This change in temperature to cool will be very expensive.

$$c) N_{min} = \frac{\ln \left[\left(\frac{0.99}{0.01} \right) / \left(\frac{0.017}{0.983} \right) \right]}{\ln 10.5} = 3.68 \approx 4 \text{ stages}$$

d) $\Delta V_{feed} = \sum \frac{\alpha_{i-v} \cdot F_{zi}}{\alpha_{i-v} - \phi} \rightarrow D = \frac{42.7 \cdot 7.75}{42.7 - \phi} + \frac{36 \cdot 15.5}{36 - \phi} + \frac{10.5 \cdot 23.25}{10.5 - \phi} + \frac{34.1}{1 - \phi} + \frac{0.64 \cdot 34.1}{0.64 - \phi} + \frac{0.41 \cdot 40.3}{0.41 - \phi}$
 $\phi = 2.14 \quad (1 < \phi < 10.5)$

$$V_{min} = \sum \frac{\alpha_{i-v} (D \cdot x_{i,D})}{\alpha_{i-v} - \phi} = \frac{42.7 \cdot 7.75}{42.7 - 2.14} + \frac{36 \cdot 15.5}{36 - 2.14} + \frac{10.5 \cdot 23.25}{10.5 - 2.14} + \frac{0.5197}{1 - 2.14} = 52.96 \frac{\text{kmol}}{h}$$

$$\left(\frac{L}{D} \right)_{min} = \frac{V_{min} - D}{D} = \frac{52.96 - 46.8472}{46.8472} = 0.13$$

e) $\frac{L}{D} = M \times \left(\frac{L}{D} \right)_{min} \quad \left\{ \begin{array}{l} M = 1.04 : \frac{L}{D} = (1.04)(0.13) = 0.1352 \\ X = \frac{\frac{L}{D} - \left(\frac{L}{D} \right)_{min}}{\frac{L}{D} + 1} = \frac{0.1352 - 0.13}{0.1352 + 1} = 0.0046 \end{array} \right.$

$M = 1.10$

$\frac{L}{D} = (1.10)(0.13) = 0.143$

$X = \frac{\frac{L}{D} - \left(\frac{L}{D} \right)_{min}}{\frac{L}{D} + 1} = \frac{0.143 - 0.13}{0.143 + 1} = 0.0114$

$Y = \frac{1 - X^{0.0031}}{1 - 0.9935 X^{0.0031}} = 0.721$

$N = \frac{(Y + N_{min})}{1 - Y} = \frac{0.721 + 4}{1 - 0.721} = 16.92 \approx 17 \text{ stages}$

$Y = \frac{1 - X^{0.0031}}{1 - 0.9935 X^{0.0031}} = 0.7$

$N = \left(\frac{Y + N_{min}}{1 - Y} \right) = 15.7 \rightarrow 16 \text{ stages}$

$M = 2 ; \frac{L}{D} = (2)(0.13) = 0.26$

$X = \frac{0.26 - 0.13}{0.26 + 1} = 0.103$

$Y = \frac{1 - X^{0.0031}}{1 - 0.9935 X^{0.0031}} = 0.53$

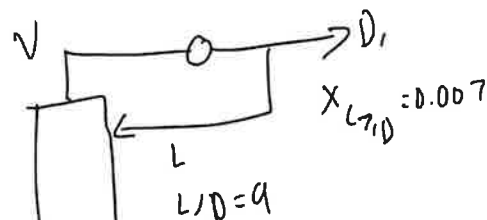
$N = \frac{(Y + N_{min})}{1 - Y} = 9.64 \rightarrow 10 \text{ stages}$

f) The distillation would not be reasonably economical and an alternative could be cheaper since it needs a lot of stages.

D9. We are separating a mixture of benzene, toluene, and xylene in a distillation-rectifying column. The column has a total condenser and no reboiler. The feed is a saturated vapor that is fed into the bottom stage of the column, flow rate $F = 150 \text{ kmol/h}$, and feed is 52.0 mol % benzene, 38.5 mol % toluene, and remainder xylenes. Pressure is 1.0 atm. CMO is valid, and the relative volatilities are constant: $\alpha_{\text{Ben-Tol}} = 2.22$, $\alpha_{\text{Tol-Xy}} = 2.01$. The column is at 1.0 atm. The reflux ratio $L/D = 9$, and the distillate is 0.007 mole fraction toluene.

The column is at 1.0 atm. The reflux ratio $L/D = 9$, and the distillate is 0.007 mole fraction toluene.

- Based on the best assumption you can make, use mass balances and CMO to calculate: B, mole fractions in bottoms, D, and mole fractions in distillate.
- Although the column has a feed and bottoms removal, we can still operate at total reflux ($D = 0$ so that $L/V = 1$). At total reflux, how many stages are required to obtain the separation achieved in part a?
- Use the Fenske equation to estimate xylene mole fraction in the distillate.
- What is the minimum reflux ratio for separation in part a, but with xylene distillate mole fraction from part c?
- Use the Gilliland correlation to estimate the actual number of stages if $L/D = 9$.



$$F = 150 \frac{\text{kmol}}{\text{h}}$$

$$\begin{aligned} z_{C6} &= 0.52 \text{ (B)} \text{ LH} \\ z_{C7} &= 0.385 \text{ (T)} \text{ HH} \\ z_{C8} &= 0.095 \text{ (X)} \text{ HH} \\ (1 - 0.385 - 0.52) &= 0.095 \end{aligned}$$

$$\sum z_i = 1$$

$$F = D + B \quad Fz = Dx_D + Bx_B$$

a) assume no HH in distillate:

$$x_{C8,D} = 0$$

$$x_{C6,D} = 1 - 0.007 = 0.993$$

$$(x_{C7,D})$$

(assume no LH in bottoms $x_{C6,B} \approx 0$)

$$L_6 = Fz_{C6} - D x_{C6,D} + B x_{C6,B}$$

$$(150)(0.52) = D(0.993) \rightarrow D = 78.55 \frac{\text{kmol}}{\text{h}}$$

$$F = D + B \rightarrow 150 = 78.55 + B \rightarrow B = 71.45 \frac{\text{kmol}}{\text{h}}$$

$$Fz_{C8} = D x_{C8,D} + B x_{C8,B}$$

$$(150)(0.095) = (78.55)(0.007) + 71.45 x_{C8,B}$$

$$x_{C8,B} = 0.19$$

$$x_{C7,B} = 1 - 0.19 = 0.81$$

$$b) N_{\min} = \frac{\ln \left[\left(\frac{x_{C6}}{x_{C7}} \right)_D / \left(\frac{x_{C6}}{x_{C7}} \right)_B \right]}{\ln \alpha_{\text{LH-HH}}}{B \text{ T}} = \frac{\ln \left[\left(\frac{0.993}{0.007} \right) / \left(\frac{0.52}{0.385} \right) \right]}{\ln 2.22}$$

$$N_{\min} = 18.6 \approx 19 \text{ stages}$$

$$c) N_{\min} = \frac{\ln \left[\left(\frac{x_{C7}}{x_{C8}} \right)_D / \left(\frac{x_{C7}}{x_{C8}} \right)_B \right]}{\ln \alpha_{\text{HH-HH}}}{T \text{ X}} = \frac{\ln \left[\left(\frac{0.007}{0.19} \right) / \left(\frac{0.385}{0.81} \right) \right]}{\ln 2.01} = 19$$

$$x_{C8,D} = \frac{0.007}{\left(\frac{0.19}{0.81} \right)^{1/19} e^{1/2.01}} = 5.11 \times 10^{-8}$$

$$d) \Delta V_{\text{feed}} = F(1-q) = \sum \frac{\alpha_{i-v} \cdot F \cdot z_i}{\alpha_{i-v} - \phi} \rightarrow 150 = \frac{2.22 \cdot 78}{2.22 - \phi} + \frac{1 \cdot 14.25}{1 - \phi} + \frac{0.498 \cdot 57.75}{0.498 - \phi}$$

$$150 = \frac{173.16}{2.22 - \phi} + \frac{14.25}{1 - \phi} + \frac{28.7545}{0.498 - \phi}$$

$$150(2.22 - \phi)(1 - \phi)(0.498 - \phi) =$$

$$173.16(1 - \phi)(0.498 - \phi) + 14.25(2.22 - \phi)(0.498 - \phi) + 28.7545(2.22 - \phi)(1 - \phi)$$

$$\phi = 1.4$$

$$V_{\min} = \frac{2.22 \cdot 78}{2.22 - 1.4} + \frac{0.54985}{1 - 1.4} + \frac{0.498 \cdot 4 \times 10^{-6}}{0.498 - 1.4} = 214.32 \frac{\text{kmol}}{\text{h}}$$

$$\left(\frac{L}{D} \right)_{\min} = \frac{214.32 - 78.55}{78.55} = 1.7$$

$$e) Y = \frac{q - 1.73}{q + 1} = 0.73$$

$$Y = \frac{1 - 0.73^{0.0031}}{1 - 0.4935(0.73)^{0.0071}} = 0.13$$

$$N = \frac{0.13 + 19}{1 - 0.13} = 21.99$$

$\sim 22 \text{ stages}$

D13.* We have a column separating benzene, toluene, and cumene. The column has a total condenser, a total reboiler, and nine equilibrium stages. The feed is 25.0 mol% benzene, 30.0 mol% toluene, and 45.0 mol% cumene. Feed rate is 100 mol/h, and the feed is a saturated liquid at 1.0 atm. The equilibrium data can be represented as constant relative volatilities: $\alpha_{BT} = 2.5$, $\alpha_{TT} = 1.0$, and

$\alpha_{CT} = 0.21$. We desire 99.0% recovery of toluene in the distillate and 98.0% recovery of cumene in the bottoms.

a. Determine the required external reflux ratio.

b. If $\alpha_{BT} = 2.25$ instead of 2.5, what is the value of L/D?

$$a) F R_{i,D} = \frac{D \cdot x_{i,D}}{F \cdot z_i} \quad \begin{cases} D \cdot x_{B,D} = (100 \cdot 0.25) = 25 \frac{\text{kmol}}{\text{h}} \\ D \cdot x_{T,D} = 0.99(100 \cdot 0.3) = 29.7 \frac{\text{kmol}}{\text{h}} \\ D \cdot x_{C,D} = (1 - 0.99)(100 \cdot 0.45) = 0.9 \frac{\text{kmol}}{\text{h}} \end{cases}$$

$$D = \sum (D \cdot x_{i,D}) = 25 + 29.7 + 0.9 = 55.6 \frac{\text{kmol}}{\text{h}}$$

$$D = \frac{2.5 \cdot 29.7}{2.5 - \phi} + \frac{1 \cdot 0.9}{1 - \phi} + \frac{0.21 \cdot 25}{0.21 - \phi}$$

$$D = \frac{74.25}{2.5 - \phi} + \frac{0.9}{1 - \phi} + \frac{5.25}{0.21 - \phi} \rightarrow 0 = 74.25(1 - \phi)(0.21 - \phi) + 0.9(2.5 - \phi)(0.21 - \phi) + 5.25(2.5 - \phi)(1 - \phi)$$

$$0 = 15.5925 - 74.25\phi - 15.5295\phi + 74.25\phi^2 + 0.4225 - 2.25\phi - 0.189\phi + 0.9\phi^2 + 13.125 - 13.125\phi - 5.25\phi + 5.25\phi^2$$

$$0 = 80.4\phi^2 - 110.65\phi + 29.19 \quad \phi = 1.58$$

$$V_{min} = \frac{2.5 \cdot 29.7}{2.5 - 1.58} + \frac{1 \cdot 0.9}{1 - 1.58} + \frac{0.21 \cdot 25}{0.21 - 1.58} = 75.32$$

$$\left(\frac{L}{D}\right)_{min} = \frac{75.32 - 55.6}{55.6} = 0.355 \quad N_{min} = \frac{\ln \left(\frac{0.99}{1 - 0.99} \right) / \left(\frac{1 - 0.98}{0.98} \right)}{\ln \left(\frac{1}{0.21} \right)} \approx 6 \text{ stages}$$

$$Y = \frac{N - N_{min}}{N + 1} = \frac{9 - 6}{10} = 0.3 \quad X = \left(\frac{1 - 0.3}{1 - 0.99(0.3)} \right)^{0.4031} = 0.41$$

$$0.41 = \frac{\frac{L}{D} - 0.355}{\frac{L}{D} + 1} \rightarrow 0.41 \frac{L}{D} + 0.41 = \frac{L}{D} - 0.355 \rightarrow \frac{L}{D} = 1.3$$

$$b) 0 = \frac{2.25 \cdot 29.7}{2.25 - \phi} + \frac{0.9}{1 - \phi} + \frac{5.25}{0.21 - \phi} \rightarrow 0 = 14.0325 - 66.825\phi - 14.0325\phi + 66.825\phi^2 + 0.4225 - 2.025\phi - 0.189\phi + 0.9\phi^2 + 11.8125 - 11.8125\phi - 5.25\phi + 5.25\phi^2$$

$$V_{min} = \frac{66.825}{2.25 - 1.51} + \frac{0.9}{1 - 1.51} + \frac{5.25}{0.21 - 1.51} = 84.5$$

$$\left(\frac{L}{D}\right)_{min} = \frac{84.5 - 55.6}{55.6} = 0.52 \quad X = 0.41 \rightarrow 0.41 = \frac{\frac{L}{D} - 0.52}{\frac{L}{D} + 1} \rightarrow 0.41 \frac{L}{D} + 0.41 = \frac{L}{D} - 0.52$$

